

On the Representational Geometry of Dynamic Programs

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Abstract

Standard neural architectures often fail to generalize to longer inputs for dynamic programming (DP) targets. We investigate what makes this hard geometrically. Every finite min-plus DP is a shortest path on a DAG, which is equivalently a tropical polynomial whose extended Newton polyhedron encodes the decision boundary of which path wins. We prove these three descriptions (graph, polynomial, polyhedron) form isomorphic semirings at two levels — formal polynomials and their computed functions — connected by operations that characterize all structural redundancies. We then address the length-generalization question geometrically: does the decision boundary at length T decide the boundary at $T + 1$? We present two structural negatives. The semiring’s two native ways to reduce dimension (setting a variable to each identity) are neither injective nor always closed within the DP. Series and parallel composition fail to construct all DAG topologies from smaller sub-DAGs, and even all terminal-only operations do not capture all DP compositions.

1 Introduction

A dynamic program is defined by a recurrence that does not depend on the size of its input, so a model that has truly learned the recurrence would run at any length. Standard sequence architectures do not. Trained on combinatorial optimization targets, they fit the training lengths and then degrade sharply beyond them, a failure documented across recurrent and attention-based models and robust to scale [Delétang et al., 2023, Anil et al., 2022]. Tropical Attention [Hashemi et al., 2025] is a conspicuous exception; replacing addition and multiplication with max-plus (tropical) operations ($\max, +$), it achieves strong length out-of-distribution (OOD) generalization on classic DP problems.

The premise is classical. After unrolling, every finite min-plus dynamic program is: (1) a single-pair shortest-path problem on a decision DAG with a single source s and single sink z , which we call *two-terminal* [Bellman, 1957, Gondran and Minoux, 1979]; and (2) a tropical polynomial [Joswig, 2021]. For alignment with the shortest-path formulation, we work throughout in the dual $(\min, +)$ semiring $\mathbb{T} = \mathbb{R} \cup \{+\infty\}$, equivalent to $(\max, +)$ under negation [Joswig, 2021]. We write $a \oplus b = \min(a, b)$ and $a \odot b = a + b$ when emphasizing the semiring structure.

Our work is focused on length generalization: a learner observes small graphs with at most T variables and must predict every graph after that. We therefore investigate whether this is possible via the questions: (1) What algebraic structure in DP allows algorithms to work on every length? (2) Does this structure allow us to infer larger instances from smaller ones?

Composing two-terminal DAGs in series adds their path lengths; composing them in parallel takes the minimum (Figure 3 in Appendix A). We first formalize the relationship between graphs, polynomials, and geometry via a triple semiring isomorphism (Section 2). Multiple *formal* polynomials may map to the same *function* — $\min(0, x, 2x) = \min(0, 2x)$ for all x despite different supports — so we construct the formalism at both levels and relate them with congruence quotients. Graph structures are even richer than their polynomials, so we provide a canonical *deshared* normal form for every tropical polynomial and a list of invariant operations that traverse equivalence classes on DAGs (Section 3).

A natural way to relate larger instances to smaller ones is substitution at the semiring identities: setting a variable to $+\infty$ removes it from a shortest-path or knapsack instance, while setting it to 0 removes it from a min-subarray or assignment instance. We employ the above framework to provide a characterization of length recovery under substitution in terms of graph, polynomial, polyhedron, and the *decision boundary* $\mathcal{T}(f)$, known as the *tropical hypersurface* [Zhang et al., 2018]. We show in Section 4 that both substitutions are not always closed within the DP, and do not recover length even when they are: the T -th instance does not determine the $(T+1)$ -th. Finally, Section 5 identifies series and parallel composition as the two faces of Bellman’s equation on decision DAGs, then exposes a two-layered topological insufficiency: together they do not generate all topologies of terminal-only operations, and terminal-only operations do not capture all DP compositions.

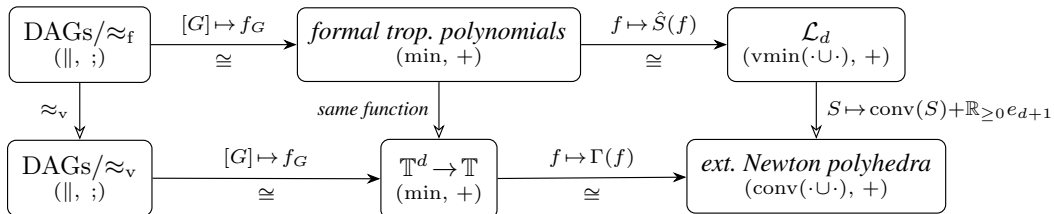
2 The triality: one problem, three languages

Appendix B details the tropical geometry background used throughout. A *formal tropical polynomial* in d variables is a finite tropical sum of monomials, $f(x) = \bigoplus_{\alpha} c_{\alpha} \odot x^{\odot \alpha} = \min_{\alpha} (c_{\alpha} + \langle \alpha, x \rangle)$, with exponents $\alpha \in \mathbb{Z}_{\geq 0}^d$ and coefficients $c_{\alpha} \in \mathbb{T}$ ($c_{\alpha} = +\infty$ meaning the term is absent). Its *support* is $S(f) = \{\alpha : c_{\alpha} \neq +\infty\}$, and its *lifted support* $\hat{S}(f) = \{(\alpha, c_{\alpha}) : \alpha \in S(f)\}$ raises each point to its coefficient height. Terms sharing an exponent keep only the lesser coefficient, $(c \oplus c') \odot x^{\odot \alpha}$; call this per-exponent reduction *vmin* (vertical minimum), and say f is in *normal form* once reduced by it, one coefficient per exponent. The normal-form lifted supports $\mathcal{L}_d$ — finite subsets of $\mathbb{Z}_{\geq 0}^d \times \mathbb{R}$ with at most one point over each exponent — form a commutative idempotent semiring under $\text{vmin}(\cdot \cup \cdot)$ and Minkowski addition (Proposition 9). The *extended Newton polyhedron* is $\Gamma(f) = \text{conv}(\hat{S}(f)) + \mathbb{R}_{\geq 0} e_{d+1}$, the convex hull of the lifted support extended upward. Two polynomials compute the same function if and only if their extended Newton polyhedra agree [Joswig, 2021]. Thus, convexity is the second quotient, and *vmin* becomes *conv*. The ordinary *Newton polytope* $\mathcal{N}(f) = \text{conv}(S(f))$ is its vertical projection, constructed via monomials but not their coefficients.

Every two-terminal weighted DAG G computes a min-plus *path polynomial*. This may be constructed by enumerating paths π from source s to sink z as follows: $f_G = \min_{\pi: s \rightarrow z} \sum_{e \in \pi} w_e$, the minimum over s – z paths in normal form. For DAGs A and B , $f_{A;B} = f_A + f_B$ and $f_{A \parallel B} = \min(f_A, f_B)$ (Proposition 14). However, two graphs may have different topologies but compute exactly the same shortest-path function, so we define equivalence relations that erase exactly this excess: $G \approx_f G'$ iff $\hat{S}(f_G) = \hat{S}(f_{G'})$, and $G \approx_v G'$ iff $f_G, f_{G'}$ compute the same function.

Both are congruences (compatible with $;$ and $\parallel$). The following theorem says each quotient produces a semiring isomorphic to its associated polynomial and polyhedron: at each level, graph, algebra, and geometry are the *same* semiring.

Theorem 1. *The following diagram of commutative idempotent semirings commutes; horizontal maps are isomorphisms, vertical maps are surjective homomorphisms.*



The proof is in Appendix D. The practical consequence is that a claim may be formulated and checked in whichever of the three languages makes it easiest, then translated to the others.

3 A complete calculus of invariant operations on DAG equivalences

We now give the operations relating two $\approx$ -equivalent graphs. The *deshared normal form* $D_f(G)$ is the parallel bundle of chains giving each monomial of f_G its own s - z path, edges in variable order: the graph-language canonical form of the polynomial as a min of monomials. Mediant deletion (dropping dominated chains) then gives the *function normal form* $D_v(G)$. Four local operations, each realizing a semiring property (Table 1 in Appendix E), carry any graph to these forms.

Theorem 2. *For all two-terminal DAGs G, G' :*

- (i) *Reduction. A finite sequence of formal moves carries G to $D_f(G)$, using one vertex split per path intersection (Lemma 16). Mediant deletions (Lemma 18) then carry $D_f(G)$ to $D_v(G)$.*
- (ii) *Canonicity. $D_f(G) = D_f(G')$ iff $G \approx_f G'$, and $D_v(G) = D_v(G')$ iff $G \approx_v G'$; the chains of $D_v(G)$ are exactly the vertices of $\Gamma(f_G)$. Both congruences are therefore decidable.*

Connectivity follows: two graphs are joined by formal moves iff they are $\approx_f$, and by formal moves plus mediant deletions iff they are $\approx_v$, since each side can reduce to its normal form. The complete argument is in Appendix E.

4 Substitution geometry

To relate lengths we need an operation on congruence-classes that is well-defined across all three languages and reduces the variable count. The semiring offers two identity substitutions: write P_i for $x_i \mapsto +\infty$ (the min-identity) and Q_i for $x_i \mapsto 0$ (the +-identity). Over a field, two slices and subtraction recover affine dependence on a coordinate; tropically there is no subtraction, so a slice can only restrict. Does the restriction suffice? We present two mechanisms of identification failure.

Proposition 3. *Let f_G be a tropical polynomial in $t + 1$ variables. Writing π_i for deletion of coordinate i and $\hat{\pi}_i$ for its height-preserving lift, each of P_i and Q_i acts by a single transformation across the three languages, and $Q_i(f_G) \leq P_i(f_G)$ pointwise (a face lies inside the shadow):*

	P_i : coordinate face	Q_i : coordinate shadow
<i>algebra</i>	the $\alpha_i=0$ monomials, coefficients intact	support $\pi_i(S(f_G))$, $c_\beta = \min\{c_\alpha : \pi_i(\alpha)=\beta\}$
<i>polyhedron</i>	$\Gamma(f_G) \cap \{\alpha_i=0\}$, a coordinate face	$\hat{\pi}_i(\Gamma(f_G))$, the projection along α_i
<i>hypersurface</i>	recession complex in direction e_i	slice at $x_i=0$
<i>graph</i>	delete every x_i -edge: $[G \setminus E_i]$	reweight every x_i -edge to 0, then merge

The two graph rows carry the two routes of the opening paragraph: for P_i , “delete every x_i -edge” descends to classes; for Q_i , “reweight every x_i -edge to 0, then merge” is a rule on the deshared canonical form, since contraction on a general graph can create or destroy an s - z path. The hypersurface row reads the same operations on the tropical hypersurface $\mathcal{T}(f_G)$, the decision boundary of the DP (Section 1; Zhang et al., 2018): P_i restricts to the strategies that ignore position i , while Q_i slices through the boundary at $x_i = 0$. Neither substitution is injective, and occasionally they are not *closed*; in Figure 1, neither P_i nor Q_i sends the proper-subset instance to a proper-subset instance. The exact fiber is computed in Corollary 21 (Appendix F).

5 A hierarchy beyond series and parallel

Since substitution is insufficient, cross-length structure must come from operations that *build* a longer instance from shorter ones. Consider Bellman’s equation of optimal substructure, $f_{t+1}(x) = \min_d(w_d + f_t^{(d)}(x))$. Series and parallel composition correspond to two ways of modifying a Bellman expression at time t : extending the time step and updating the prior, respectively. We show that these operations are doubly insufficient to express the topology of DP instances.

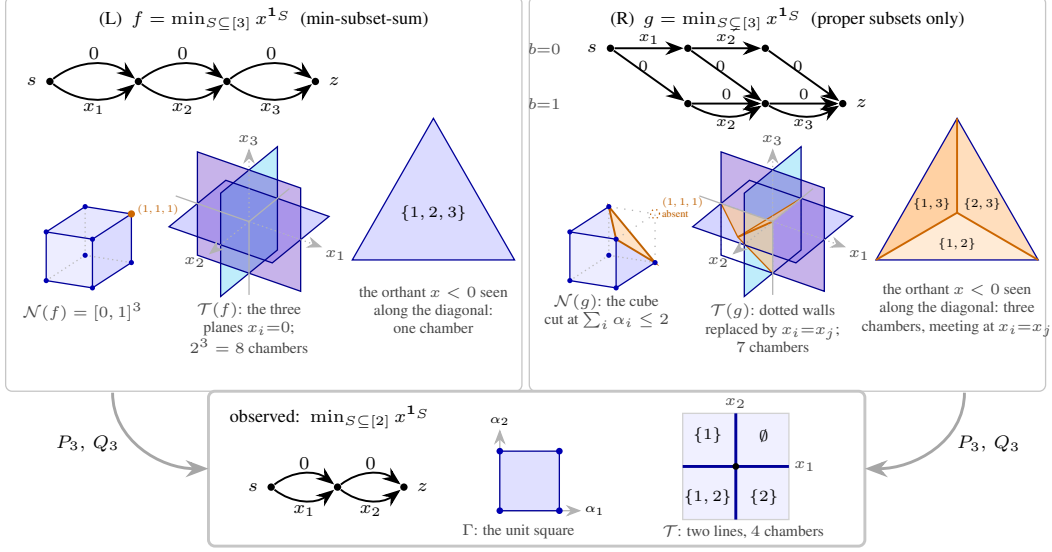


Figure 1: Two length-3 problems that no substitution distinguishes. **(L)** min-subset-sum, support $\{0, 1\}^3$; **(R)** the same over proper subsets, support $\{0, 1\}^3 \setminus \{(1, 1, 1)\}$. Middle panels show the decision boundaries (positive half-axes behind, dotted). The all-ones monomial is off the coordinate face and redundant under the projection, so both P_3 and Q_3 , applied to either problem, yield the same length-2 object (bottom).

Series-parallel $\subsetneq$ terminal-only. A terminal-only operation may be thought of as an *edge substitution* Φ_H on a template (unweighted) DAG H , replacing each edge with a sub-DAG (Appendix G). Series and parallel composition are the two simplest edge substitutions (templates with one or two edges), but they do not generate all edge-substitution topologies: the witness is the Wheatstone graph W (Figure 2, left).

Proposition 4. *The Wheatstone edge substitution Φ_W (Figure 2) is not generated by series and parallel composition since W is non-series-parallel [Duffin, 1965, Valdes et al., 1982].*

Terminal-only $\subsetneq$ all DP compositions. Clamped min-subarray (Figure 2, right) composes two instances by gluing $s_A = s_B$, $z_A = z_B$, and the last internal vertex of A to the first of B , so that a subarray may cross the boundary — an identification of *internal* vertices.

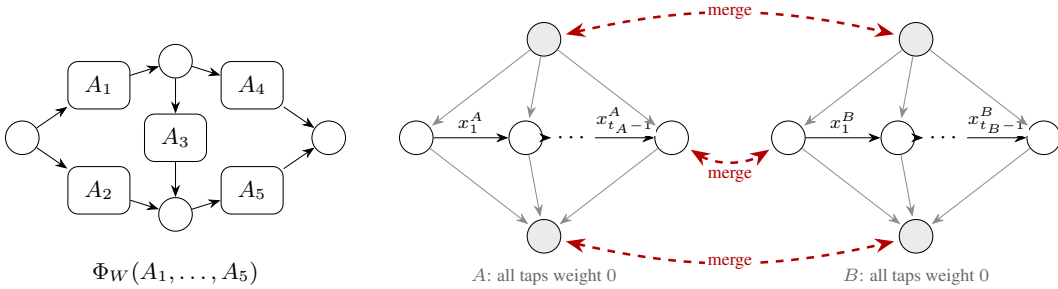


Figure 2: **(L)** The Wheatstone edge substitution $\Phi_W(A_1, \dots, A_5)$. **(R)** Merging two instances of clamped min-subarray: the two sources, the two sinks, and the last internal vertex of A with the first internal vertex of B .

6 Conclusion

The triality (Theorem 1) means that compositional structure discovered in any one of the three languages — graph, algebra, polyhedron — transfers exactly to the others. The invariant calculus

then characterizes the full redundancy in the DAG encoding, identifying which structural differences are artifacts of the representation rather than intrinsic properties of the computed function. The negative results constrain equivariant architecture design. Semiring-native substitutions cannot lift the polyhedral geometry from length T to length $T+1$ (Section 4), and the compositional symmetries available to Bellman-style reasoning are doubly incomplete: series-parallel does not generate all DAG topologies, and terminal-only operations do not capture all DP compositions (Section 5). Together these bound what $(\min, +)$ structure alone in an architecture can achieve for length generalization, and point toward richer compositional operations that would be needed.

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A Series and parallel composition

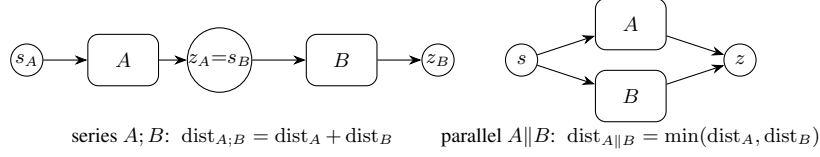


Figure 3: Series and parallel composition of two-terminal DAGs: series identifies z_A with s_B , parallel identifies the two sources and the two sinks.

B Tropical geometry in a nutshell

This appendix collects the definitions and examples needed to read the main text. We follow the conventions of Joswig [Joswig, 2021] and Maclagan–Sturmfels [Maclagan and Sturmfels, 2015]; proofs and formal statements are in Appendix C.

B.1 The tropical semiring

The *tropical semiring* is the set $\mathbb{T} = \mathbb{R} \cup \{+\infty\}$ equipped with the operations: *tropical addition* $a \oplus b := \min(a, b)$ and *tropical multiplication* $a \odot b := a + b$. The additive identity is $+\infty$ ($\min(a, +\infty) = a$) and the multiplicative identity is 0 ($a + 0 = a$). Every finite element has a multiplicative inverse ($a \odot (-a) = 0$), but tropical addition is *idempotent*: $a \oplus a = a$, so no finite element has an additive inverse and $\mathbb{T}$ is a semiring, not a ring. The $(\max, +)$ convention is equivalent under negation; we use $(\min, +)$ throughout, aligned with shortest-path formulations.

B.2 Tropical polynomials

A *tropical monomial* $c \odot x_1^{\odot \alpha_1} \odot \cdots \odot x_d^{\odot \alpha_d}$ (with $\alpha \in \mathbb{Z}_{\geq 0}^d$, $c \in \mathbb{R}$) evaluates to the affine function $c + \alpha_1 x_1 + \cdots + \alpha_d x_d = c + \langle \alpha, x \rangle$, since tropical exponentiation is repeated addition: $x^{\odot k} = kx$.

A *tropical polynomial* is a finite tropical sum of monomials:

$$f(x) = \bigoplus_{\alpha \in S} c_\alpha \odot x^{\odot \alpha} = \min_{\alpha} (c_\alpha + \langle \alpha, x \rangle).$$

Its graph is the lower envelope of finitely many affine functions — piecewise linear and concave.

Example 5 (A univariate tropical polynomial). Let $f(x) = 4 \oplus (2 \odot x) \oplus (1 \odot x^{\odot 2}) \oplus (3 \odot x^{\odot 3}) = \min(4, 2+x, 1+2x, 3+3x)$. Its graph is the lower envelope of four lines (Figure 4). The envelope bends at $x = -2, 1, 2$; this is the *tropical vanishing set* (the points where two or more monomials tie for the minimum).

Formal vs. function. Two formal polynomials can evaluate to the same function: for instance, $\min(0, x, 2x) = \min(0, 2x)$ for all x , because the monomial x is never the unique minimizer. This formal/function distinction is what the two rows of the triality (Theorem 1) capture.

Support and lifted support. The *support* $S(f) = \{\alpha : c_\alpha \neq +\infty\}$ records which monomials are present. The *lifted support* $\hat{S}(f) = \{(\alpha, c_\alpha) : \alpha \in S(f)\} \subset \mathbb{R}^d \times \mathbb{R}$ raises each exponent to its coefficient height. When two terms share an exponent, their tropical sum keeps the lesser coefficient; this per-exponent reduction is called *vmin* (vertical minimum), and a polynomial in *normal form* has been reduced by *vmin*.

B.3 The geometric dictionary

The dome. The region weakly below the graph, $\mathcal{D}(f) = \{(x, s) \in \mathbb{R}^d \times \mathbb{R} : s \leq f(x)\}$, is a convex polyhedron cut out by one half-space per monomial. Each facet of $\mathcal{D}(f)$ is the locus where one monomial is the unique minimizer; the function bends exactly at the ridges (codimension-2 faces) of $\mathcal{D}(f)$.

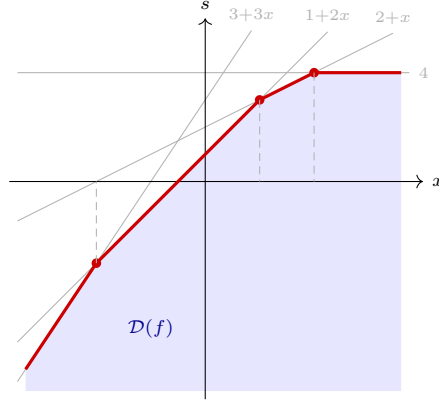


Figure 4: The dome $\mathcal{D}(f)$ of $f = \min(4, 2+x, 1+2x, 3+3x)$ from Example 5. Each facet is a monomial's locus of dominance; the three vertices of the dome project to the tropical zeroes $\{-2, 1, 2\}$.

Newton polytope and extended Newton polyhedron. The *Newton polytope* $\mathcal{N}(f) = \text{conv}(S(f)) \subseteq \mathbb{R}^d$ records which monomials are present, forgetting coefficients. The *extended Newton polyhedron*

$$\Gamma(f) = \text{conv}(\hat{S}(f)) + \mathbb{R}_{\geq 0} e_{d+1}$$

retains the coefficients: it is the convex hull of the lifted support, extended upward by an infinite ray. Its lower faces encode the polynomial's evaluation, and the key fact is that $\Gamma(f) = \Gamma(g)$ if and only if f and g agree as functions [Joswig, 2021, Thm. 1.13].

Tropical hypersurface. The *tropical hypersurface*

$$\mathcal{T}(f) = \{x \in \mathbb{R}^d : \text{the minimum in } f(x) \text{ is attained by } \geq 2 \text{ monomials}\}$$

is the locus where strategies tie for optimality, i.e. a decision boundary. It is a polyhedral complex of pure dimension $d - 1$, the projection of the codimension-2 skeleton of the dome.

Example 6 (The tropical line). The polynomial $f(x, y) = 0 \oplus x \oplus y = \min(0, x, y)$ has three monomials. The tropical hypersurface $\mathcal{T}(f)$ consists of three rays emanating from the origin (Figure 5): $\{x = 0 \leq y\}$, $\{y = 0 \leq x\}$, and $\{x = y \leq 0\}$. Each ray separates two regions of linearity, one per monomial that dominates there.

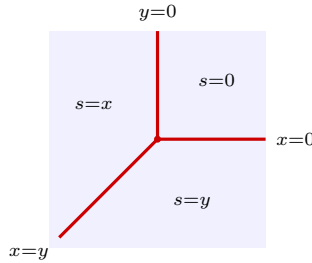


Figure 5: The tropical line of Example 6: the hypersurface of $f = \min(0, x, y)$ is three rays meeting at the origin, separating the three regions on which a single monomial attains the minimum.

B.4 Two-terminal DAGs and path polynomials

A *two-terminal DAG* is an acyclic directed graph with a designated source s and sink z such that every vertex lies on some s – z path. Its *path polynomial* is

$$f_G(x) = \bigoplus_{\pi : s \rightarrow z} \bigodot_{e \in \pi} w_e = \min_{\pi : s \rightarrow z} \sum_{e \in \pi} w_e,$$

the tropical sum over all s – z paths of the tropical product of edge weights. Each path contributes one monomial; the polynomial is their tropical sum.

Series composition ($A; B$, identifying $z_A = s_B$) gives $f_{A;B} = f_A \odot f_B = f_A + f_B$; parallel composition ($A \parallel B$, identifying sources and sinks) gives $f_{A \parallel B} = f_A \oplus f_B = \min(f_A, f_B)$. Thus $G \mapsto f_G$ preserves both $\parallel$ and $;$ from two-terminal DAGs to tropical polynomials — the starting point of the triality in Section 2.

C Polynomials, configurations, and polyhedra

Definition 7. For a formal tropical polynomial $f = \min_{\alpha} (c_{\alpha} + x^{\alpha})$ in d variables in normal form (one least coefficient per exponent), the support is $S(f) = \{\alpha : c_{\alpha} \neq +\infty\}$ and the lifted configuration is $\hat{S}(f) = \{(\alpha, c_{\alpha}) : \alpha \in S(f)\} \subseteq \mathbb{R}^d \times \mathbb{R}$.

Definition 8. For finite $S \subseteq \mathbb{Z}_{\geq 0}^d \times \mathbb{R}$, $\text{vmin}(S)$ keeps over each exponent occurring in S only the least point above it; thus $\text{vmin}(S) = S$ iff S has at most one point over each exponent.

Proposition 9. Let $\mathcal{L}_d$ be the finite subsets of $\mathbb{Z}_{\geq 0}^d \times \mathbb{R}$ with at most one point over each exponent, with addition $\text{vmin}(S \cup T)$ and multiplication $\text{vmin}\{(\alpha + \alpha', c_{\alpha} + c_{\alpha'})\}$. Then $\mathcal{L}_d$ is a commutative idempotent semiring (identities $\emptyset$ and $\{(0, 0)\}$), and $f \mapsto \hat{S}(f)$ is a semiring isomorphism from normal-form formal tropical polynomials onto $\mathcal{L}_d$.

Proof. Formal tropical Laurent polynomials are a semiring [Joswig, 2021, Observation 2.15]; nonnegative-exponent configurations form a sub-semiring containing $\emptyset$ and $\{(0, 0)\}$, and $f \mapsto \hat{S}(f)$ is a relabelling carrying tropical min to vmin of the union and $+$ to vmin of the sumset. $\square$

Definition 10. For f with support S , $\Gamma(f) = \text{conv}\{(\alpha, c_{\alpha}) : \alpha \in S\} + (\{0\} \times \mathbb{R}_{\geq 0}) \subseteq \mathbb{R}^d \times \mathbb{R}$. Its lower faces (outer normal $(y, 1)$) project onto $\mathcal{N}(f) = \text{conv}(S)$ [Joswig, 2021, §1.1].

Lemma 11. (i) $\Gamma(f)$ is its lower hull plus the upward ray, so the lower hull determines it. (ii) $\Gamma(f) \mapsto f$ is a bijection from extended Newton polyhedra onto tropical polynomial functions. [Joswig, 2021, Observation 1.8, Theorem 1.13]

Lemma 12. On $\mathcal{L}_d$ set $S \sim T$ iff $\text{conv}(S) + \mathbb{R}_{\geq 0}e_{d+1} = \text{conv}(T) + \mathbb{R}_{\geq 0}e_{d+1}$. Then $\sim$ is a congruence for $(\min, +)$, and $\gamma : S \mapsto \text{conv}(S) + \mathbb{R}_{\geq 0}e_{d+1}$ induces a semiring isomorphism $\mathcal{L}_d / \sim \xrightarrow{\cong} (\{\Gamma\}, \text{conv}(\cdot \cup \cdot), +)$.

Proof of Lemma 12. Write $R = \mathbb{R}_{\geq 0}e_{d+1}$. Each Γ satisfies $\Gamma + R = \Gamma$, hence is determined by its lower support function — the map $x \mapsto \min_{(\alpha, c) \in \Gamma} (\langle \alpha, x \rangle + c)$, which is exactly the tropical polynomial function carried by the configuration. The lower support function of a convex hull of a union is the pointwise minimum of the two [Rockafellar, 1970, Cor. 16.5.1], i.e. $\min$; the support function of a Minkowski sum is the sum [Rockafellar, 1970, Thm. 13.3], i.e. $+$. A point discarded by vmin lies on R above a retained point and is invisible to γ , so γ intertwines the operations; that this holds forces $\sim$ to be a congruence. $\gamma(\emptyset) = \emptyset$, $\gamma(\{(0, 0)\}) = R$ are the identities. $\square$

Lemma 13. Let $\varphi : R \rightarrow R'$ be a semiring isomorphism and θ, θ' congruences with $a \theta b \iff \varphi(a) \theta' \varphi(b)$. Then there is a unique $\bar{\varphi} : R/\theta \rightarrow R'/\theta'$ with $\bar{\varphi} \circ q = q' \circ \varphi$, and it is a semiring isomorphism.

Proof. We must show $\bar{\varphi}([a]) := [\varphi(a)]_{\theta'}$ is a well-defined semiring isomorphism $R/\theta \rightarrow R'/\theta'$.

Well-defined. If $[a]_{\theta} = [b]_{\theta}$, then $a \theta b$, so by hypothesis $\varphi(a) \theta' \varphi(b)$, hence $[\varphi(a)]_{\theta'} = [\varphi(b)]_{\theta'}$.

Injective. If $\bar{\varphi}([a]) = \bar{\varphi}([b])$, i.e. $\varphi(a) \theta' \varphi(b)$, the reverse implication gives $a \theta b$, so $[a] = [b]$.

Surjective. For any $[r'] \in R'/\theta'$, surjectivity of φ gives a with $\varphi(a) = r'$, so $\bar{\varphi}([a]) = [r']$.

Homomorphism. For either operation $*$ (with $*$ ' its counterpart in R'):

$\bar{\varphi}([a] * [b]) = \bar{\varphi}([a * b]) = [\varphi(a * b)]_{\theta'} = [\varphi(a) *' \varphi(b)]_{\theta'} = [\varphi(a)]_{\theta'} *' [\varphi(b)]_{\theta'} = \bar{\varphi}([a]) *' \bar{\varphi}([b])$, using that φ is a homomorphism (third equality) and that θ' is a congruence (fourth equality). The identity elements map correctly since φ preserves them.

Uniqueness. Any ψ with $\psi \circ q = q' \circ \varphi$ satisfies $\psi([a]) = q'(\varphi(a)) = [\varphi(a)]_{\theta'} = \bar{\varphi}([a])$. $\square$

Proposition 14. *For two-terminal DAGs, $f_{A;B} = f_A + f_B$ and $f_{A\|B} = \min(f_A, f_B)$; hence the two congruences $\approx_f, \approx_v$ are compatible with $;$ and $\|$.*

Proof. The s - z paths of $A;B$ are concatenations of an A -path and a B -path, weight the sum, so distributivity gives $f_A + f_B$; those of $A\|B$ are the disjoint union, so the min splits as $\min(f_A, f_B)$. Normal form commutes with both. $\square$

D Proof of the triality (Theorem 1)

Proof. Write $\Phi : [G] \mapsto f_G$.

(1) *Formal row.* Since $\approx_f$ is a congruence (Proposition 14), $\|$ and $;$ are well-defined on classes.

- *Injective:* by the definition of $\approx_f$.
- *Operation-preserving:* by Proposition 14.
- *Identities:* the single $+\infty$ -edge maps to $\emptyset$, the single 0-edge to $\{(0, 0)\}$.
- *Surjective:* given normal-form f , build one chain per support monomial — α_i edges labelled x_i and one edge of weight c_{α} — and place the chains in parallel. Its path polynomial is f .

So Φ is an isomorphism onto the formal polynomials, a semiring by Proposition 9; composing with $f \mapsto \hat{S}(f)$ gives the chain.

(2, 3) *Function row.* Two graphs compute the same function iff their polynomials do iff their extended Newton polyhedra agree, so the three congruences match under the formal-row isomorphisms and the descent lemma applies. Concretely, let $\theta_G, \theta_F, \theta_L$ be, respectively:

- $[G]_f \theta_G [H]_f$ iff $G \approx_v H$;
- $f \theta_F g$ iff they evaluate to the same function;
- $\sim$ of Lemma 12.

Now $G \approx_v H$ iff f_G, f_H evaluate equally, so θ_G and θ_F correspond under $[G] \mapsto f_G$; and $f \theta_F g$ iff $\Gamma(f) = \Gamma(g)$ (Lemma 11(ii)) iff $\hat{S}(f) \sim \hat{S}(g)$, so θ_F and θ_L correspond. θ_L is a congruence (Lemma 12), hence so are the others by transport. Applying Lemma 13 to the two isomorphisms of (1) with these matched congruences yields the function-row isomorphisms, and since the vertical maps are the quotient maps, both squares commute. No operation on Γ is recomputed. $\square$

E Invariance: normal form and completeness

Definition 15. *A two-terminal DAG is deshared if distinct s - z paths meet only at s and z (a parallel bundle of chains). $D_f(G)$ has one canonicalized chain per support monomial of f_G ; $D_v(G)$ deletes additionally every chain whose lifted point lies weakly above the lower hull of the others. Both may be constructed directly from the path polynomial itself, so DAGs G, G' are equal exactly when their path polynomials are equal.*

Lemma 16. *For a two-terminal DAG G , with $P(v)$ the number of s - z paths through internal vertex v , put $\Pi(G) = \sum_v (P(v) - 1)$. Then (1) G is deshared iff $\Pi(G) = 0$; (2) a vertex split is available iff $\Pi(G) > 0$; (3) exactly $\Pi(G)$ splits carry G to deshared form.*

Proof. By two-terminality each internal v has $P(v) \geq 1$.

- (1) $\Pi(G) = 0$ iff every $P(v) = 1$ iff every internal vertex has in- and out-degree 1 on the path subgraph, which is desharedness.
- (2) $\Pi(G) > 0$ iff some vertex has in- or out-degree > 1 , which admits a split.

- (3) A split of v into v_1, v_2 has $P(v_1) + P(v_2) = P(v)$ and changes no other $P(v')$, so

$$\Pi \text{ drops by } (P(v) - 1) - (P(v_1) - 1) - (P(v_2) - 1) = 1;$$

with (2), exactly $\Pi(G)$ splits reach deshared form.

□

Proposition 17. *The following preserve $\approx_f$: (1) series-run reweighting — permuting and contracting the edges within a chain, preserving its sum; (2) parallel-bundle bookkeeping — reordering parallel branches, combining two with equal exponent by min of coefficients (idempotency when equal), and deleting or adjoining a branch dominated at its own exponent; (3) tropical-zero adjunction — adding or removing a parallel $+\infty$ -edge or a subdividing 0-edge; (4) vertex splitting and merging (Figure 6).*

Proof. Write $\mathcal{P}(G)$ for the multiset of s - z path monomials. Since f_G is the min over $\mathcal{P}(G)$ in normal form, it suffices to show each operation leaves $\mathcal{P}(G)$ unchanged up to dominated entries.

- (1) The run is unbranched, so every path through it picks up the $+$ -product of the run's weights; paths avoiding it are untouched.
- (2) Permuting branches permutes $\mathcal{P}(G)$. Deletion: the surviving branch A at the same exponent with $c_A \leq c_C$ already attains the min-least coefficient there.
- (3) A path through a $+\infty$ -edge has constant $+\infty$ and is dominated or unsupported; deletion reads the argument backwards.
- (4) Out-split at w along $\text{out}(w) = O_1 \sqcup O_2$; in-split is symmetric. *Acyclicity*: a cycle of G' through w_j maps to a cycle of G through w . *Path bijection*: an s - z path of G meeting w , entering along $e \in \text{in}(w)$ and leaving along $e' \in O_j$, lifts to an s - z path of G' through w_j of the same weight; conversely, a path of G' meets at most one of w_1, w_2 (else a cycle exists), and identifying them recovers a path of G . So $\mathcal{P}(G') = \mathcal{P}(G)$. Merging is the inverse.

□

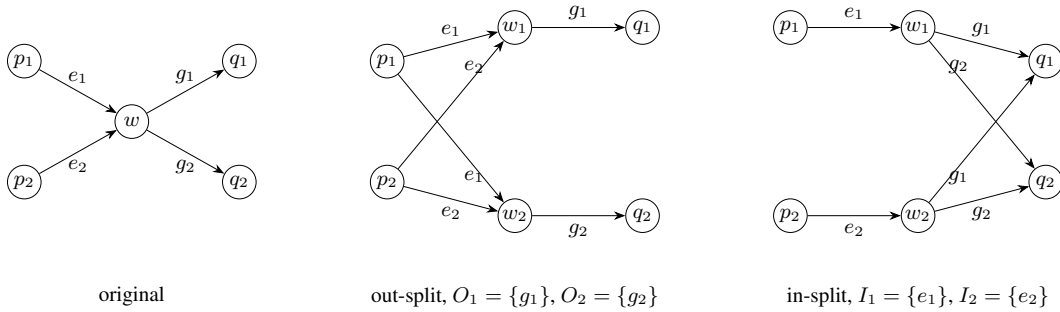


Figure 6: The two vertex splits, realizing the distributivity clause of Proposition 17. Left: an internal vertex w with in-edges e_1, e_2 and out-edges g_1, g_2 . Middle: the out-split along $O_1 = \{g_1\}$, $O_2 = \{g_2\}$, with each of w_1, w_2 carrying its own copy of e_1 and e_2 . Right: the in-split along $I_1 = \{e_1\}$, $I_2 = \{e_2\}$, with each of w_1, w_2 carrying its own copy of g_1 and g_2 . In both panels every $p_i \rightarrow q_j$ path of the original survives exactly once.

Lemma 18. *On a bundle of chains with pairwise distinct exponents and no $+\infty$ -edge, deleting a chain preserves the computed function iff its lifted point (α, c_α) lies weakly above the lower hull of the others, iff it is not a vertex of Γ , iff it is nowhere the unique minimizer (Figure 7); and at most $d + 1$ chains ($d = \dim \mathcal{N}(f)$) are needed. The forward direction (a certificate $\Rightarrow$ deletion preserves the function) is unconditional.*

Proof. Write $f_C(x) = \langle \alpha_C, x \rangle + c_C$ and $f_{A_i}(x) = \langle \alpha_i, x \rangle + c_i$; set $S' = \{(\alpha_i, c_i)\}_{i \leq r}$.

Certificate $\Rightarrow$ deletion (unconditional). A certificate gives $N\alpha_C = \sum_i k_i \alpha_i$ and $Nc_C \geq \sum_i k_i c_i$, so for every x ,

$$Nf_C(x) = \langle N\alpha_C, x \rangle + Nc_C \geq \sum_i k_i f_{A_i}(x) \geq N \min_j f_{A_j}(x).$$

Hence $f_C \geq \min_j f_{A_j}$ pointwise, and a branch that never strictly beats the others may be dropped from a min without changing it.

Equivalences (under distinctness and no $+\infty$).

- *Not a vertex $\Leftrightarrow$ nowhere the unique minimizer.* $\Gamma(f_B)$ recedes in the e_{d+1} direction, so every supporting functional with positive last coordinate scales to $(\alpha, c) \mapsto \langle \alpha, x \rangle + c$ for some x ; its minimum on $\Gamma(f_B)$ is the bundle's value at x , attained uniquely at (α_C, c_C) iff C is the unique minimizer there.
- *Nowhere unique minimizer $\Leftrightarrow$ deletion preserves the function.* The value of the bundle changes at x iff C is the unique minimizer at x .
- *Weakly above the lower hull $\Leftrightarrow$ not a vertex.* $\Gamma(f_B) = \text{conv}(S' \cup \{(\alpha_C, c_C)\}) + \mathbb{R}_{\geq 0} e_{d+1}$. If $(\alpha_C, c_C) \in \Gamma(S')$ it is a convex combination of points other than itself, hence not extreme; conversely, if not extreme it lies in $\Gamma(S')$ because all other extreme points belong to S' .

Support bound. Weakly above the lower hull means $\alpha_C \in \text{conv}(\{\alpha_i\})$ with $c_C \geq h(\alpha_C)$, where h is the lower-hull function of S' . By Carathéodory, α_C lies in a simplex of at most $d+1$ vertices of $\text{conv}(\{\alpha_i\})$, with rational barycentric coordinates $\lambda_i \geq 0$, $\sum \lambda_i = 1$, and $\sum \lambda_i c_i = h(\alpha_C) \leq c_C$. Clearing the common denominator N gives $k_i = N\lambda_i \in \mathbb{N}$: a certificate supported on at most $d+1$ chains. $\square$

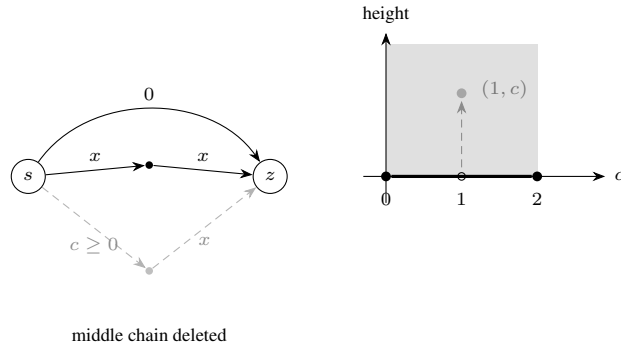


Figure 7: Mediant deletion (Lemma 18) on the family $G_c = 0 \parallel (x; x) \parallel (c; x)$, computing $\min(0, c + x, 2x)$. Left: the middle chain (weight $c \geq 0$ then x) is deleted for every $c \geq 0$, leaving $\min(0, 2x)$. Right: the lifted points and extended Newton polyhedron.

Proof of Theorem 2. (i) *Reduction.* By Lemma 16, $\Pi(G)$ vertex splits deshare G . Within each chain, Proposition 17(1) sorts the variable edges and contracts the trailing constants to one edge; a $+\infty$ -chain contracts to a single $+\infty$ -edge and is deleted by (3). Two chains sharing an exponent have equal variable prefixes and differ only in a final constant; merging their prefix vertices (inverse out-splits) leaves a parallel pair $c \parallel c'$, replaced by $\min(c, c')$ via (2) (idempotency deletes exact duplicates). What remains is one chain per support monomial, i.e. $D_f(G)$. For the mediant pass, the vertices of $\Gamma(f_G)$ are fixed and every non-vertex chain fails Lemma 18, so is deletable against the vertices; performing all such deletions leaves $D_v(G)$.

- (ii) *Canonicity.* $D_f(G)$ is a function of f_G alone (Definition 15), so $D_f(G) = D_f(G')$ iff $f_G = f_{G'}$, i.e. $\approx_f$. The mediant pass keeps exactly the vertices of Γ , and two functions agree iff their polyhedra, hence vertex sets, agree (Lemma 11(ii)), giving the $\approx_v$ statement and decidability.

- (iii) *Connectivity*. Each formal move is invertible (merging inverts splitting, adjunction inverts deletion, etc.), so the reduction path of (i) can be reversed. Two graphs G, G' with the same canonical form are therefore connected by $G \rightarrow D(G) = D(G') \rightarrow G'$; by (ii), same canonical form is equivalent to congruence. Conversely each move preserves the class. $\square$

Table 1: The calculus of invariant operations on DAGs derived from tropical semiring properties.

graph operation	tropical semiring property
<i>series-run reweighting</i> : redistribute weights along an unbranched run, sum preserved	$+$ associative, commutative
<i>parallel-bundle bookkeeping</i> : permute or delete/adjoin a dominated branch	min assoc., comm., idempotent
$+\infty$ -edge adjunction: add/remove $+\infty$ -weighted edges (preserving 2-terminality)	$\min(a, +\infty) = a, a + \infty = +\infty$
<i>vertex splitting / merging</i> : split edges at an internal vertex into two copies, or merge	distributivity of $+$ over min
<i>mediant deletion</i> : delete a path whose monomial lies weakly above the lower hull of the others	redundant monomials may be deleted by convexity

F Geometry of the two substitutions

Definition 19. Write $\pi_i : \mathbb{R}^d \rightarrow \mathbb{R}^{d-1}$ for deletion of coordinate i , $\alpha_{-i} = \pi_i(\alpha)$, and $\hat{\pi}_i = \pi_i \times \text{id}_{\mathbb{R}}$ for its lift keeping the height. The shadow uses $\hat{\pi}_i(\Gamma(f))$, which retains coefficients; $\pi_i(\mathcal{N}(f))$ forgets them.

Definition 20. For a polyhedral complex $\mathcal{C}$ and $v \neq 0$, $\text{rec}_v(\mathcal{C}) = \{C \in \mathcal{C} : v \in \text{rec}(C)\}$ is the subcomplex of cells stable under translation by v [Joswig, 2021, §A].

Proof of Proposition 3. P_i acts by a single transformation across the three languages by the triality (Theorem 1): the rule “set $x_i = +\infty$ ” respects $\approx_f$, hence descends to classes and has one avatar in each language.

Algebra. With $\hat{S} = \hat{S}(f_G)$, the sign of α_i partitions $\hat{S} = \hat{S}_i^> \sqcup \hat{S}_i^0$ ($\alpha_i > 0$ resp. $= 0$). Since $+\infty$ is absorbing for $+$, setting $x_i = +\infty$ kills every monomial with $\alpha_i \geq 1$ and fixes the rest with coefficients, so $\hat{S}(P_i(f_G)) = \hat{S}_i^0$; writing h for the $\hat{S}_i^>$ block, $f_G = \min(h, P_i(f_G))$.

Geometry. The exponents lie in $\mathbb{Z}_{\geq 0}^{t+1}$ and the upward ray fixes coordinate i , so $\Gamma(f_G) \subseteq \{\alpha_i \geq 0\}$; hence $\{\alpha_i = 0\}$ is supporting and $\Gamma(f_G) \cap \{\alpha_i = 0\}$ is a (possibly empty) face, carrying exactly $\hat{S}_i^0$. For the hypersurface, take f_G in function form. A cell of $\mathcal{T}(f_G)$ has an argmin set σ ; those with $\sigma \subseteq \hat{S}_i^0$ are stable under increasing x_i (no x_i appears), while any σ containing a monomial with $\alpha_i > 0$ is eventually dominated by a surviving one, so leaves the cell. Thus the cells stable in direction e_i are exactly those on $\hat{S}_i^0$, i.e. $\text{rec}_{e_i}(\mathcal{T}(f_G))$; projecting by π_i gives $\mathcal{T}(P_i(f_G))$, provided $\hat{S}_i^0 \neq \emptyset$ (else $P_i(f_G) = +\infty$).

Graph. The s - z paths through an x_i -edge acquire weight $+\infty$ and drop out of the min; those avoiding E_i are the paths of $G \setminus E_i$ with unchanged weight, so $f_G|_{x_i=+\infty} = f_{G \setminus E_i}$.

The Q_i column. Q_i is evaluation at $x_i = 0$, a semiring homomorphism, so it acts term by term: the term at α becomes $c_\alpha + x_{-i}^{\alpha_{-i}}$, each x_i contributing the $+$ -identity 0.

Algebra. In lifted coordinates $Q_i(f_G)$ is the min over the projected set $A = \hat{\pi}_i(\hat{S}(f_G))$; as π_i need not be injective on the support, A may carry several points over one exponent, and normal form keeps the least:

$$\hat{S}(Q_i(f_G)) = \text{vmin}(A), \quad \text{support } \pi_i(S(f_G)), \quad \text{coefficient } \min\{c_\alpha : \pi_i(\alpha) = \beta\}.$$

Geometry. Adding the upward ray R absorbs the points $v_{\min}$ discards, each lying on R above a retained one, so

$$\Gamma(Q_i(f_G)) = \text{conv}(A) + R = \hat{\pi}_i(\Gamma(f_G)),$$

since $\hat{\pi}_i$ is linear and fixes R .

Hypersurface. As $Q_i(f_G)$ is f_G restricted to $\{x_i = 0\}$, a tie of $Q_i(f_G)$ is one of f_G on that slice and conversely, save that two monomials sharing a π_i -image agree on the whole slice and merge into one; their common wall fills a full-dimensional cell on which a single monomial of $Q_i(f_G)$ dominates, and removing those relative interiors is the stated correction.

Graph. Reweighting E_i to 0 substitutes $x_i = 0$ into every path monomial, and the two normalizing moves are invariant; this is reweighting rather than contraction, which can create or destroy an s - z path and so misrepresents Q_i .

$Q_i \leq P_i$. The coordinate face lies in the coordinate shadow:

$$\Gamma(P_i(f_G)) = \hat{\pi}_i(\Gamma(f_G) \cap \{\alpha_i = 0\}) \subseteq \hat{\pi}_i(\Gamma(f_G)) = \Gamma(Q_i(f_G)),$$

and since tropical addition of functions is the convex hull of the union of their polyhedra, $\min(Q_i(f_G), P_i(f_G)) = Q_i(f_G)$, i.e. $Q_i(f_G) \leq P_i(f_G)$. $\square$

Corollary 21. For a tropical polynomial g in the variables $\{x_j : j \neq i\}$,

$$P_i^{-1}(g) = \{g \oplus h : \text{every monomial of } h \text{ has } \alpha_i \geq 1\},$$

$$Q_i^{-1}(g) = \left\{ \bigoplus_{\beta \in S(g)} c_\beta^g \odot x_{-i}^{\odot \beta} \odot x_i^{\odot k_\beta} \oplus r : k_\beta \in \mathbb{N}, \pi_i(S(r)) \subseteq S(g), c_\alpha^r \geq c_{\pi_i(\alpha)}^g \right\}.$$

The shapes are opposite: P_i pins the $\alpha_i = 0$ layer and frees everything above position i , while Q_i frees the exponent k_β at position i and constrains everything above the fiber minimum.

Proof of Corollary 21. P_i -fiber. By the algebraic clause, every $f = \min(g, h)$ with each monomial of h carrying $\alpha_i \geq 1$ has $\hat{S}(f) = \hat{S}(g) \sqcup \hat{S}(h)$ and $P_i(f) = g$. Conversely any f not of that form either omits a monomial of g , forcing $S(P_i(f)) \subsetneq S(g)$, or contributes a monomial with $\alpha_i = 0$ beyond g , forcing $S(g) \subsetneq S(P_i(f))$.

Q_i -fiber. Q_i is a homomorphism, so any

$$g_k = \min_\beta (c_\beta^g + x_{-i}^\beta + k_\beta x_i)$$

has $Q_i(g_k) = g$, since each $k_\beta x_i$ evaluates to 0. For $f = \min(g_k, r)$,

$$Q_i(f) = \min(g, Q_i(r)) = g \iff Q_i(r) \geq g \text{ pointwise;}$$

by the algebraic clause that is $\pi_i(S(r)) \subseteq S(g)$ together with $c_\alpha^r \geq c_{\pi_i(\alpha)}^g$ at every monomial of r . $\square$

G Edge substitutions and the series-parallel boundary

Definition 22. For a two-terminal DAG H with r labeled edges, the edge substitution Φ_H sends two-terminal DAGs $A_1, \dots, A_r$ to the two-terminal DAG obtained by replacing edge i of H with A_i , identifying the edge's endpoints with A_i 's source and sink. Series composition is the edge substitution into a two-edge path; parallel composition is the edge substitution into a pair of parallel edges.

Definition 23. A two-terminal DAG is series-parallel (SP) if built from a single $s \rightarrow z$ edge by series and parallel compositions. The Wheatstone graph W is K_4 minus an edge, the smallest non-SP two-terminal DAG.

Theorem 24 (Duffin, 1965, Valdes et al., 1982). A two-terminal DAG is SP iff it contains no subgraph homeomorphic to W .

Proof of Proposition 4. Specializing each A_i to a single edge, a series-parallel composite of single edges is SP (Definition 23); but Φ_W on five single edges is W , non-SP by Theorem 24. $\square$